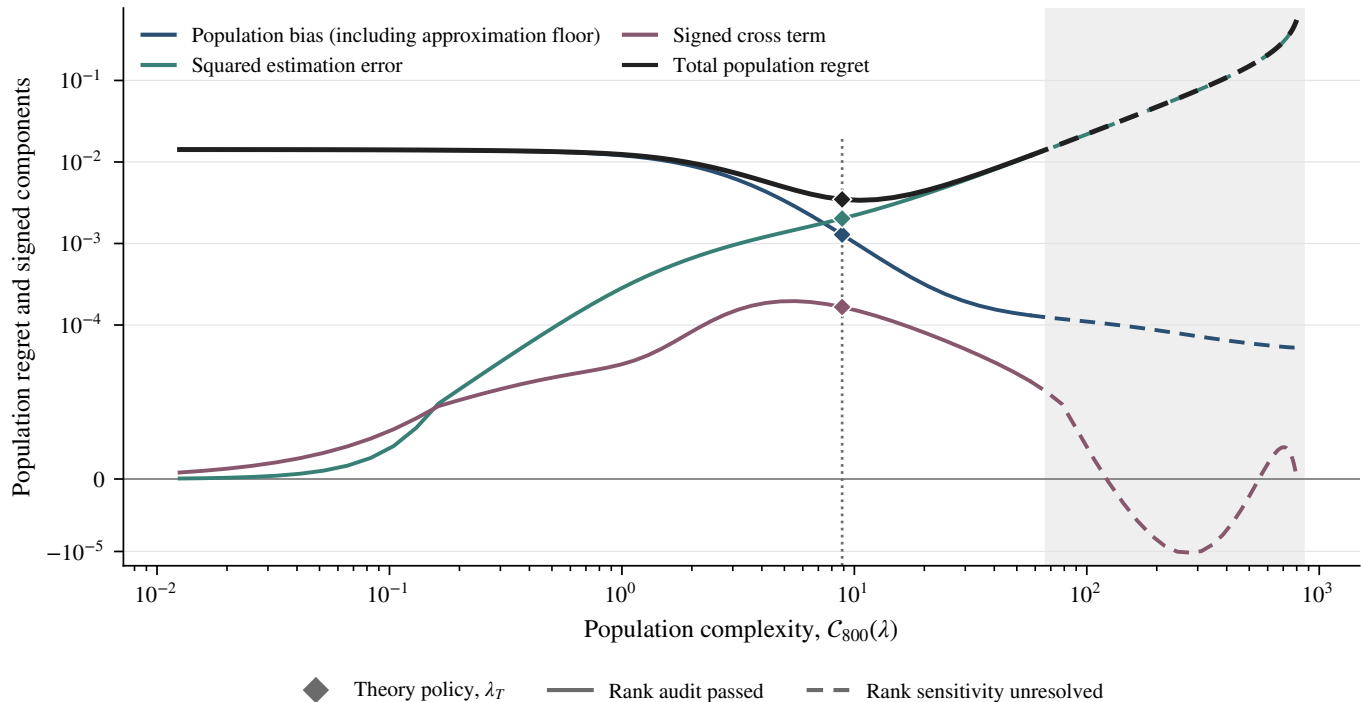


(a) Exact regret decomposition,  $T = 1440$



Signed cross term retained. Symmetric-log scale, linear within  $\pm 10^{-5}$ .